Optimization of Hydraulic Neural Networks

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Abstract

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1. Introduction

2. Problem description

This work considers the steady-state simulation of hydraulic networks, where the objective is to control the flow distribution by adjusting the cross-sectional areas of the pipes. The Hydraulic Neural Network (HNN) is modeled as a system of interconnected pipes, governed by the conservation of mass and momentum, assuming incompressible, one-dimensional flow.

2.1. Flow equations

The behavior of each pipe is described by the momentum and continuity equations. The steady-state momentum equation, accounting for frictional losses, is expressed as:

$$0 = -\nabla p - \frac{f_D}{2d_h}\rho u|u| + \rho\mathbf{F} \tag{1}$$

where p is the pressure, ρ is the fluid density, u is the mean velocity, d_h is the hydraulic diameter of the pipe, f_D is the Darcy friction factor, and $\mathbf{F}$ represents external body forces such as gravity.

The mass conservation in each pipe is represented by the continuity equation:

$$\nabla \cdot (A\rho u) = 0 \tag{2}$$

where A is the cross-sectional area of the pipe, which serves as the design variable in the optimization process.

2.2. Design variables and nonlinearity

The cross-sectional areas $\mathbf{A}$ of the pipes determine the hydraulic resistance and directly influence the pressure distribution and flow rates throughout the network. Since head losses are nonlinear functions of velocity and diameter, the relation between A and the flow rate is inherently nonlinear.

2.3. Boundary conditions

The network includes inflow and outflow boundaries, where pressures or flow rates are prescribed. Additionally, internal nodes are subject to flow continuity constraints. These boundary and connectivity conditions are essential to close the system and compute the steady-state solution.

2.4. Governing system

The full hydraulic network is thus modeled as a coupled nonlinear system of equations derived from Eqs. (1) and (2), which must be satisfied for any configuration of the design variables $\mathbf{A}$.

3. Design of Hydraulic Neural Networks

The design objective of the Hydraulic Neural Network (HNN) is to determine the optimal set of cross-sectional areas $\mathbf{A}$ for the pipes such that the flow rates at selected control nodes match predefined target values.

The flow rates at the nodes are nonlinear functions of the pipe areas $\mathbf{A}$ and the nodal pressures P, which are implicitly defined by the governing equations of the hydraulic system. The objective function is formulated as a least-squares problem:

$$\mathcal{J}(\mathbf{A}) = \sum_{i \in \mathcal{N}_{\text{target}}} \left(Q_i(\mathbf{A}) - Q_i^{\text{desired}} \right)^2 \tag{3}$$

where $\mathcal{N}_{\text{target}}$ is the set of control nodes, $Q_i(\mathbf{A})$ is the computed flow rate at node i, and Q_i^{desired} is the desired flow rate at that node.

The optimization is constrained by bounds on the cross-sectional areas:

$$A_e^{\min} \leq A_e \leq A_e^{\max} \quad \forall e \tag{4}$$

In addition, the steady-state hydraulic behavior must be satisfied, i.e., the solution $\mathbf{A}$ must ensure that the flow field obeys the momentum and continuity equations described in Eqs. (1) and (2).

Hence, the optimization problem can be summarized as:

$$\begin{aligned} \text{minimize:} \quad & \mathcal{J}(\mathbf{A}) \quad \text{as in Eq. (3)} \\ \text{subject to:} \quad & A_e^{\min} \leq A_e \leq A_e^{\max} \quad \forall e \\ & \text{and hydraulic equations (1)–(2) are satisfied} \end{aligned} \tag{5}$$

4. Numerical implementation

The numerical implementation of the proposed Hydraulic Neural Network (HNN) optimization framework was carried out using the commercial software COMSOL Multiphysics®. The hydraulic analysis was based on the steady-state solution of the momentum and continuity equations, as described in Section 2, for incompressible and unidirectional pipe flow.

The design variables, corresponding to the cross-sectional areas of the pipes, were defined as parametric variables and linked to the flow model. The nonlinear relationship between the areas and the resulting flow rates was captured through the governing equations within each evaluation of the objective function.

For the optimization process, the built-in IPOPT (Interior Point OPTimizer) algorithm from COMSOL's Optimization Module was used. IPOPT is a large-scale nonlinear optimization solver that supports inequality constraints and is suitable for problems such as the one defined in Eqs. (3)–(5).

A flowchart of the optimization procedure is presented in Figure **??**, illustrating the iterative coupling between hydraulic simulation and objective function evaluation. Convergence was assessed using both the relative change in the objective function and the norm of the parameter updates, with predefined tolerances to ensure stability and accuracy.

5. Results and discussion

The proposed Hydraulic Neural Network (HNN) was evaluated by performing a supervised classification task using the well-known Iris flower dataset. This dataset contains $N = 150$ samples, each described by four numerical features:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \text{sepal length} \\ \text{sepal width} \\ \text{petal length} \\ \text{petal width} \end{bmatrix} \tag{6}$$

Each input vector $\mathbf{x}$ is associated with a class label $c \in \{1, 2, 3\}$, representing the species *Setosa*, *Versicolor*, or *Virginica*.

5.1. Implementation in COMSOL

The classification was encoded by assigning three outlet nodes in the network, each corresponding to one of the possible species. For each input $\mathbf{x}$, the features were mapped into boundary conditions (e.g., inlet velocities or pressures), and the flow rates $\{Q_1, Q_2, Q_3\}$ at the outlets were used to determine the predicted class.

An example of the optimized hydraulic network topology used for classification is illustrated in Figure 1.

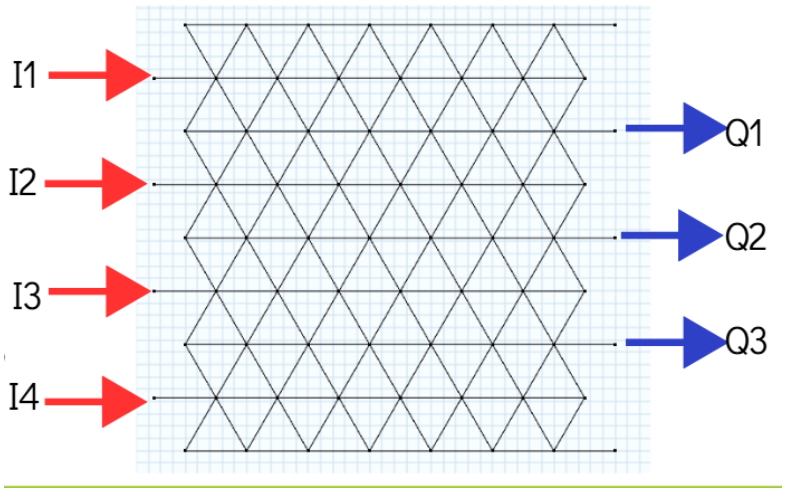


Figure 1: Classification model based on optimized hydraulic network. The outlet with highest flow rate determines the predicted class.

5.2. Training phase

A multi-objective topology optimization was performed individually for each sample in the training set. The optimization problem was defined to

maximize the flow at the outlet corresponding to the correct class and minimize the flows at the other outlets. The objective function for sample k was defined as:

$$\mathcal{J}^{(k)}(\mathbf{A}^{(k)}) = -Q_{c^{(k)}} + \lambda \sum_{i \neq c^{(k)}} Q_i \tag{7}$$

where λ is a weighting parameter, Q_i is the flow rate at outlet i, $c^{(k)}$ is the correct class of the k-th sample, and $\mathbf{A}^{(k)}$ is the vector of pipe cross-sectional areas being optimized.

The optimization was carried out using the IPOPT solver available in COMSOL's Optimization Module. The physical behavior of the system was governed by the incompressible Navier–Stokes equations in the laminar regime. Material bounds and convergence tolerances were imposed to ensure physical feasibility and numerical stability.

Figure 2 illustrates an example of a trained hydraulic network, showing the cross sectional area optimization. This structure encodes the learned decision logic through flow-routing behavior.

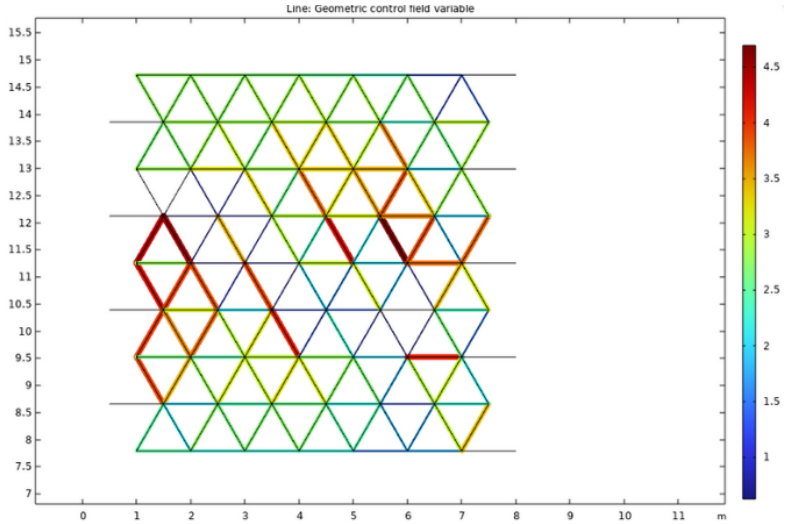


Figure 2: Trained hydraulic network topology after the optimization phase. The flow distribution encodes the classification logic across the three outlet nodes.

5.3. Testing phase

After training, the resulting pipe topology was fixed. A separate test set was then fed into the system using the same mapping of input features to

boundary conditions. For each test sample, the outlet with the highest flow rate determined the predicted class:

$$\hat{c}^{(j)} = \arg \max_{i \in \{1,2,3\}} Q_i^{(j)} \tag{8}$$

No further optimization was applied during the testing phase.

5.4. Evaluation

The classification accuracy was evaluated by comparing the predicted labels $\hat{c}^{(j)}$ against the ground truth labels $c^{(j)}$ for all test samples. The HNN successfully captured nonlinear relationships between inputs and outputs by leveraging fluid dynamics and optimized geometry, demonstrating its potential as a physically interpretable classification mechanism.

Figure 3 shows a visual representation of the Iris dataset, illustrating the separability of the three species using a combination of input features.

This result highlights the capability of HNNs to perform machine learning tasks through purely physical principles, encoded via multi-objective optimization of flow networks.

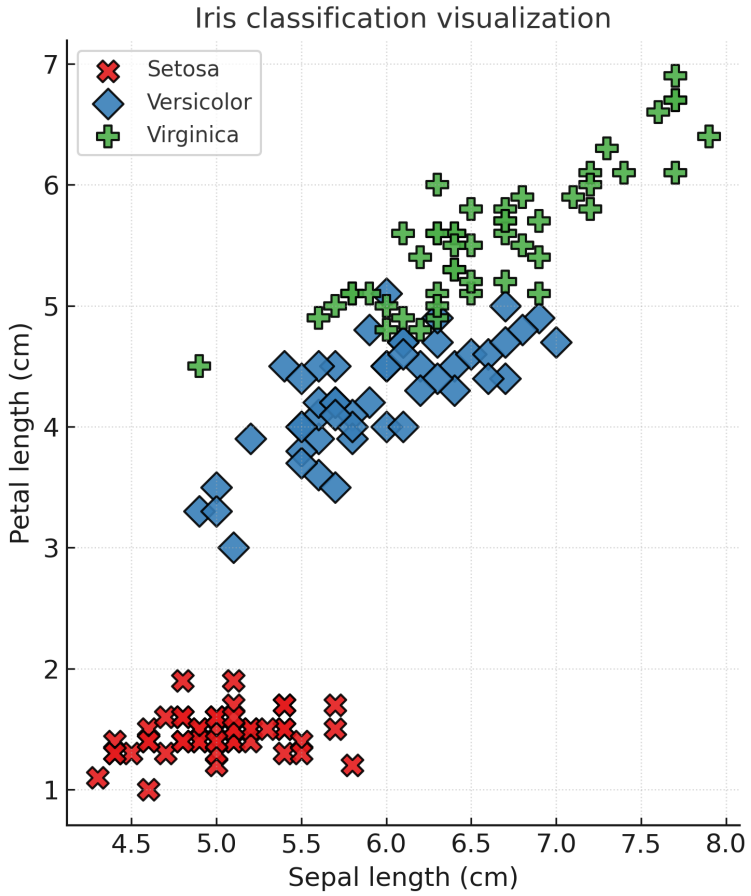


Figure 3: Scatter plot of Iris dataset using sepal length and petal length for classification. Each species is represented by a distinct marker and color.

6. Conclusions

References